[COLLEGE / INSTITUTION NAME]

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Accredited by NAAC / NBA • Affiliated to State Technological University

CampusGo

Smart Campus Front Desk & Visitor Management System

FINAL YEAR ACADEMIC PROJECT DOCUMENTATION REPORT

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ACADEMIC STREAM:

Bachelor of Engineering / Technology (B.E. / B.Tech)

ACADEMIC YEAR:

2026 – 2027

CampusGo Web Application • Powered by Next.js 16, React 19 & Salesforce Enterprise Platform

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1. Abstract

In contemporary educational institutions, front-desk visitor registration, student enquiries, faculty counseling schedules, and policy dissemination are frequently managed through fragmented manual logbooks or spreadsheets. This conventional approach results in operational latency, compromised campus security, vulnerability to identity manipulation, and poor analytics regarding daily campus footfall.

CampusGo is an enterprise-grade, centralized web application engineered specifically to digitize and modernize college front-desk administration. Built upon a high-performance frontend architecture using Next.js 16 (App Router), React 19, TypeScript, and Tailwind CSS v4, CampusGo provides an intuitive, responsive management interface designed with Razorpay-inspired visual contrast and ergonomic clarity.

Crucially, CampusGo connects seamlessly with the **Salesforce Lightning Platform CRM** backend. Rather than relying on static or mocked records, visitor registrations are validated on both client and server layers, dispatched via authenticated REST API endpoints, and committed directly to Salesforce Custom Objects (`Visitor__c`, `Student_Enquiry__c`, `Appointment__c`, and `FAQ__c`). Live footfall telemetry, real-time SOQL aggregate analytics, interactive Recharts time-series data, and an institutional knowledge base provide administrators with immediate operational intelligence.

2. Introduction & Background

Modern universities and engineering colleges host hundreds of visiting parents, academic delegates, vendor contractors, campus recruitment officers, and student applicants daily. Reception and security staff bear the responsibility of logging attendee credentials, issuing entry clearances, and routing queries to relevant departmental heads. Traditional paper logbooks exhibit critical flaws: lack of real-time auditing, departmental ticket bottlenecks, and fragmented records separated from institutional databases. CampusGo addresses these deficits by bridging a responsive web frontend with an enterprise Salesforce CRM backend.

3. Project Objectives

The primary engineering and operational objectives of the CampusGo project are:

1. **Centralized Digital Visitor Log:** Provide a rapid, structured check-in workflow capturing visitor credentials, contact numbers, host faculty, and visit purpose with automatic timestamps.
2. **Direct Salesforce CRM Persistence:** Guarantee that every visitor registration generates an authentic record in the Salesforce org (`Visitor__c`) returning a persistent 18-character Salesforce record identifier.
3. **Live Footfall & Query Telemetry:** Calculate and visualize real-time campus statistics, 7-day footfall trends, and enquiry distribution charts via live SOQL aggregation.
4. **Departmental Query & Appointment Coordination:** Streamline student enquiry tracking and faculty counseling appointment bookings with priority and status tagging.
5. **Centralized Campus Knowledge Base:** Disseminate verified institutional policies, tuition timelines, and campus guidelines from published Salesforce `FAQ__c` records.
6. **Zero-Trust Credential Security:** Implement strict server-side token management ensuring Salesforce OAuth credentials, refresh tokens, and access keys are never exposed to browser runtimes.

4. System Requirements & Specifications

4.1 Software & Development Environment

COMPONENT	SPECIFICATION / VERSION	FUNCTION
Operating System	Microsoft Windows 11 (64-bit)	Development & Testing Host Environment
Node.js Runtime	Node.js v24.19.0 / v20+ LTS	JavaScript/TypeScript Engine & Server Execution
Package Manager	npm v11.x	Dependency resolution and script management
Web Framework	Next.js 16.4.3 (App Router)	React Server Components & Server Route Handlers
UI Library	React 19.3.0 & React-DOM	Component-driven stateful user interface
Styling Engine	Tailwind CSS v4 & PostCSS 8	Utility-first CSS design tokens and responsive grid
Data Visualization	Recharts v3.10.1	Vector-based AreaChart & Donut Chart rendering
Iconography	Lucide-React v1.61.0	Feather-style SVG icons for ergonomics
Salesforce CLI	sf CLI v2.76.7 / sfdx v61.0	Salesforce DX org management & metadata deployment
Salesforce Platform	Salesforce Developer Edition (v61.0)	Enterprise CRM backend database and security model

4.2 Hardware Requirements

- **Processor:** Dual-core Intel Core i3 / AMD Ryzen 3 or higher (Quad-core recommended).
- **System Memory:** 8 GB RAM minimum (16 GB recommended for concurrent Next.js server & Salesforce CLI).
- **Storage:** 2 GB available SSD storage for repository, node_modules, and cache files.
- **Network:** Broadband Internet connectivity with HTTPS access to Salesforce endpoints.

5. Technology Stack & Frameworks

Frontend Presentation Layer

Built with **Next.js 16 App Router** and **React 19**. Employs CSS Grid layouts (`.app-layout` with 260px sidebar and fluid viewport) ensuring complete zero-overflow responsiveness across 320px mobile to 1440px desktop viewports. Uses **Recharts** for smooth SVG telemetry and **Tailwind CSS v4**.

Backend API Route Layer

Node.js server-side route handlers (`src/app/api/visitors`, `/api/dashboard/stats`, `/api/faq`, `/api/enquiries`, `/api/appointments`). Acts as a secure intermediary executing validation, token management, and authenticated Salesforce REST API dispatches.

Salesforce CRM Cloud Layer

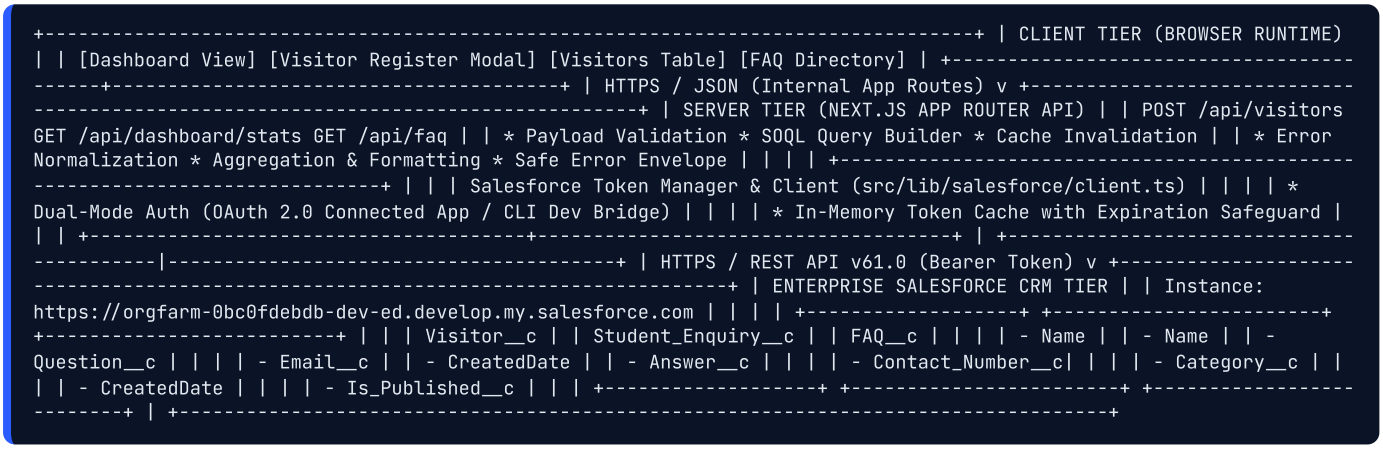
Salesforce Lightning Platform Developer Edition (Org ID: `00Dg700000LlSZREA3`). Houses custom data models (`Visitor__c`, `Student_Enquiry__c`, `Appointment__c`, `FAQ__c`) configured with field-level security, record locking, and automated timestamps.

Security & Integration Bridge

Supports dual-mode Salesforce connectivity: Connected App OAuth 2.0 Web Server / Client Credentials Flow for production environments, and a secure local development bridge leveraging the active authenticated SF CLI session with automatic token caching.

6. System Architecture & High-Level Data Flow

The interaction diagram between the CampusGo client interface, secure Next.js server proxy, and Salesforce:



7. Module Architecture & Implementation

CampusGo is structured into specialized, decoupled modules adhering to separation-of-concerns principles:

7.1 Dashboard & Operational Telemetry Module

Located at `/dashboard`, this module serves as the administrative cockpit. It initiates parallel requests to retrieve real-time counts from Salesforce: Total Visitors (`Visitor__c`), Student Enquiries (`Student_Enquiry__c`), Faculty Appointments (`Appointment__c`), and Published FAQs (`FAQ__c`). The telemetry section renders an interactive **AreaChart** for 7-day footfall trends and a segmented **Donut Chart** for category distribution.

7.2 Visitor Management & Gate Pass Module

Located at `/visitors`, this module governs logging, verification, and checkout of campus guests:

- **Live Sync Status Indicator:** Confirms active Salesforce connectivity directly in the view header.
- **Slide-Over Registration Drawer:** Validated inputs for Name, Email, Phone, Purpose, Host, Department, and Remarks.
- **Two-Way Verification:** Displays pending spinner states during REST transmission, and displays the permanent Salesforce 18-character ID (`a00...`) upon persistence confirmation.
- **Filter & Search Toolbar:** Instant client-side filtering across visitor names, hosts, and departments with status pills (`Checked In`, `Expected`, `Checked Out`).

7.3 Student Enquiries Desk Module

Located at `/enquiries`, this module facilitates academic ticketing. Student queries across categories (Admissions, Fees, Examinations, Hostels, Scholarships) are tracked with priority flags (Urgent, High, Medium, Low) and lifecycle states (Open, In Progress, Resolved, Closed).

7.4 Counseling & Appointment Schedule Module

Located at `/appointments`, this module organizes student appointments with faculty, department heads, and academic counselors with time-slot management, duration calculation, and status updating.

7.5 Institutional FAQ & Knowledge Base Module

Located at `/faq`, this module delivers authenticated institutional knowledge directly from Salesforce `FAQ__c` records with keyword debouncing, category pills, and accessible accordion expansions.

7.6 Administration & Platform Control Module

Located at `/admin`, this module presents system health metrics, direct links to the Salesforce Experience Cloud site and Lightning Web Component (LWC) apps, and synchronization diagnostics.

8. Screenshots & User Interface Walkthrough

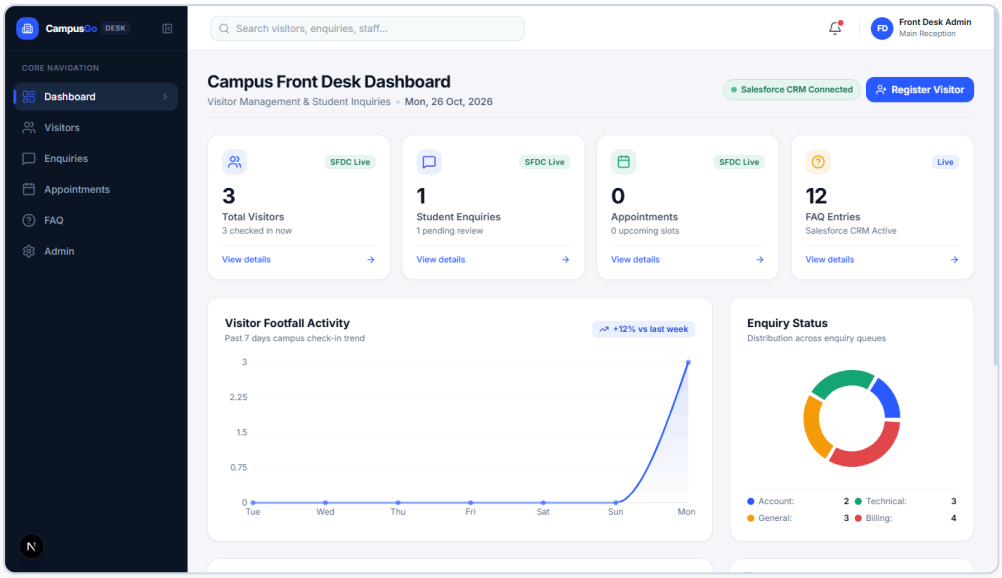


Figure 1: CampusGo Dashboard Overview & KPI Metrics

Shows the dark navy (#0B1426) navigation sidebar, top reception bar, "Salesforce CRM Connected" status pill, and four primary operational metric cards.

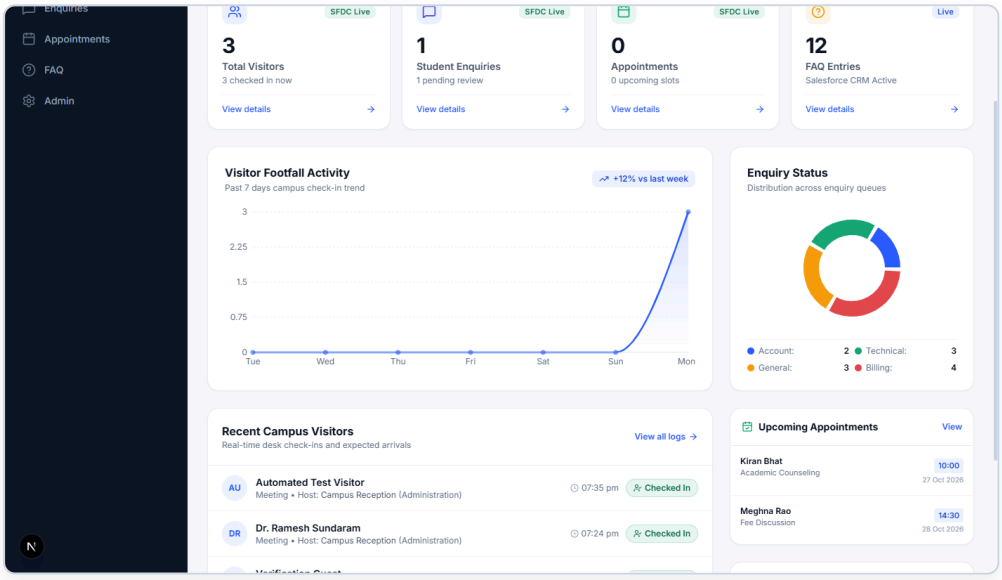


Figure 2: Real-Time Footfall Activity & Enquiry Distribution Analytics

Displays the interactive Recharts AreaChart graphing real 7-day campus footfall trends and the Donut chart showing categorized query queues.

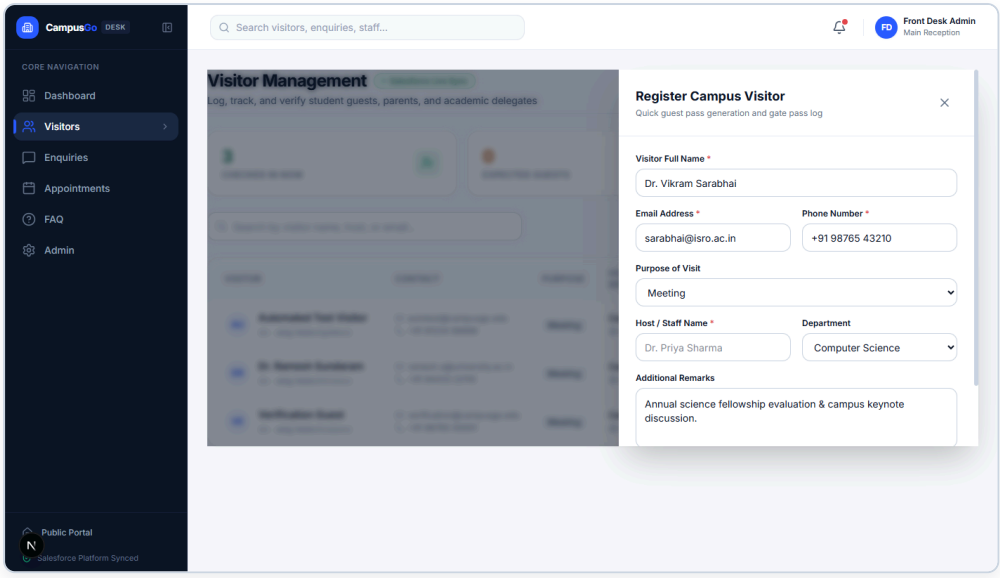


Figure 3: Campus Visitor Registration Slide-Over Form

Illustrates the validated guest pass registration interface with required input controls for Full Name, Email, Phone, Host, and Department.

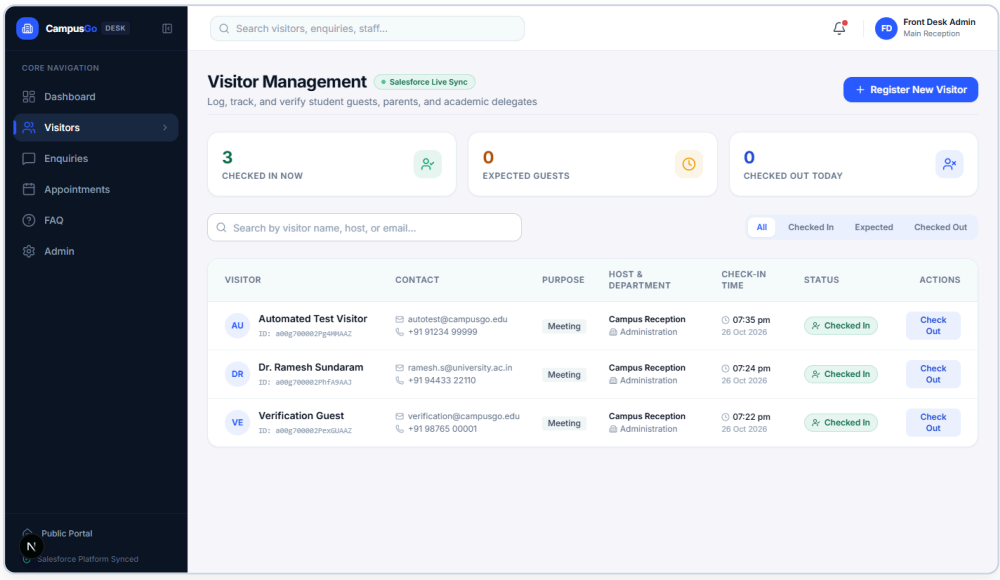


Figure 4: Real-Time Salesforce Visitor Records Directory

Displays live records queried directly from Salesforce Visitor__c (e.g. Automated Test Visitor, Dr. Ramesh Sundaram) with Salesforce Live Sync badge.

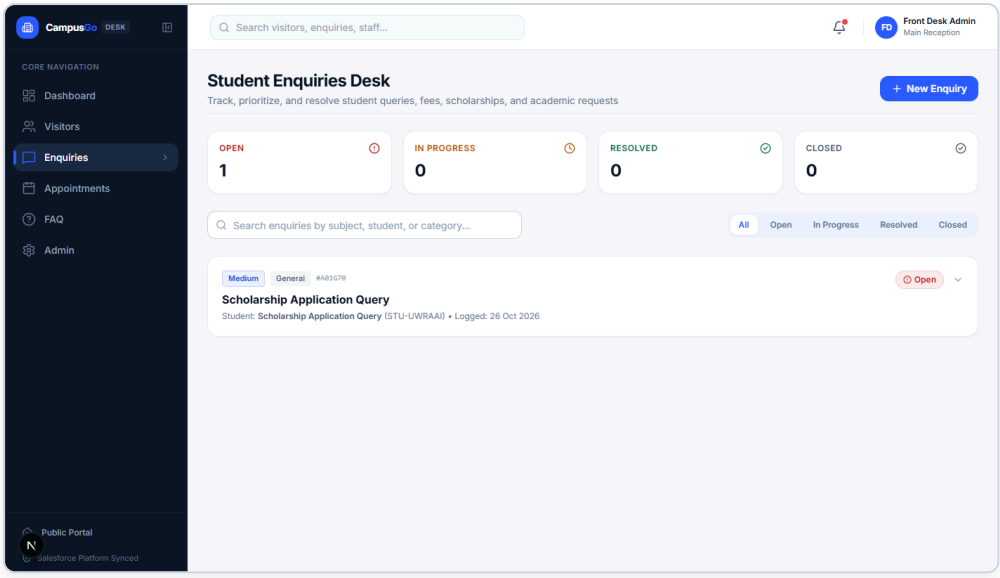


Figure 5: Student Enquiries Desk & Ticket Management

Shows enquiry queue management with color-coded priority badges (High, Urgent, Medium), category tags, and status transition controls.

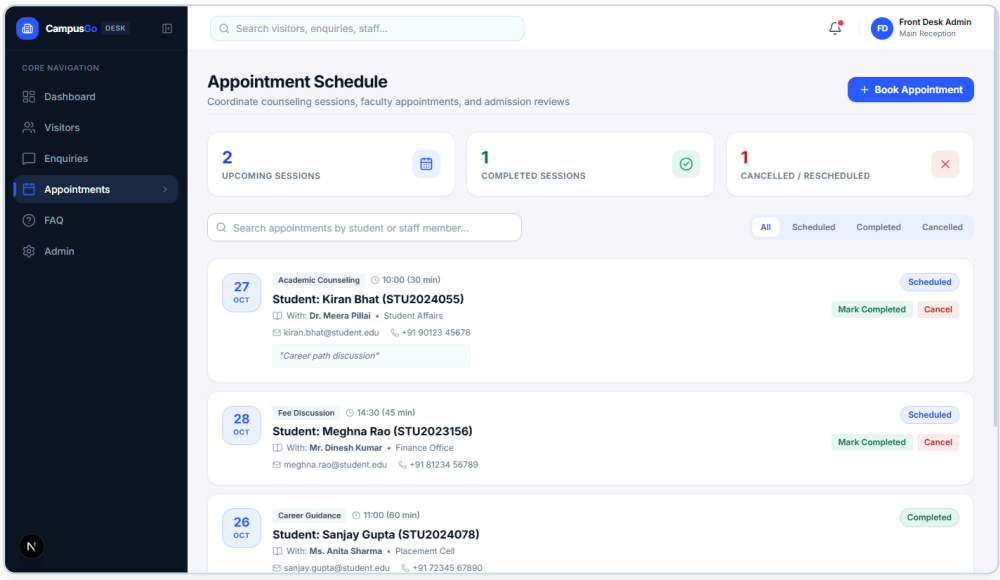


Figure 6: Faculty Counseling & Appointment Schedule

Displays scheduled academic counseling sessions, assigned faculty members, duration metrics, and date badges.

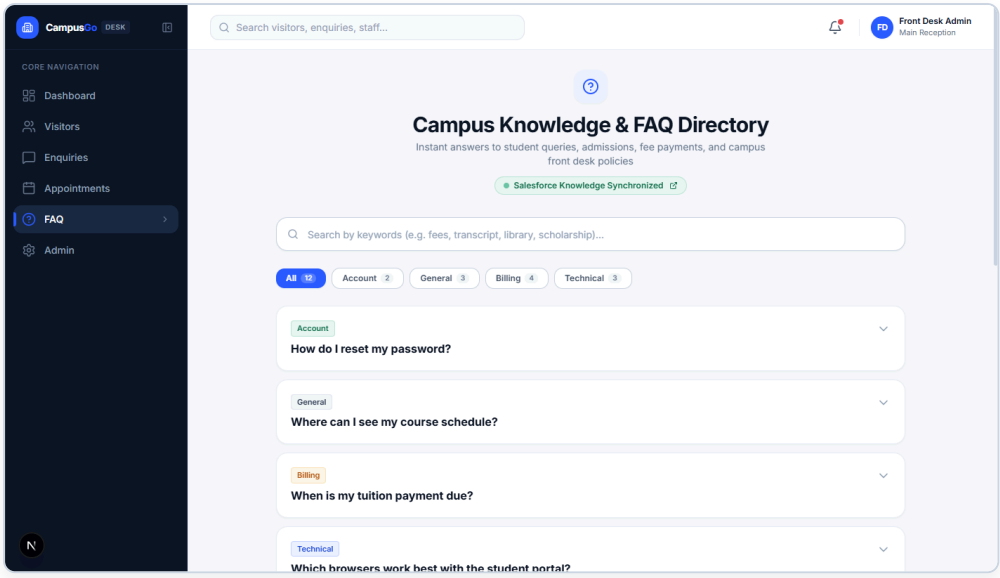


Figure 7: Campus Knowledge & FAQ Directory

Shows the synchronized institutional knowledge directory querying published articles from Salesforce FAQ__c with category filtering.

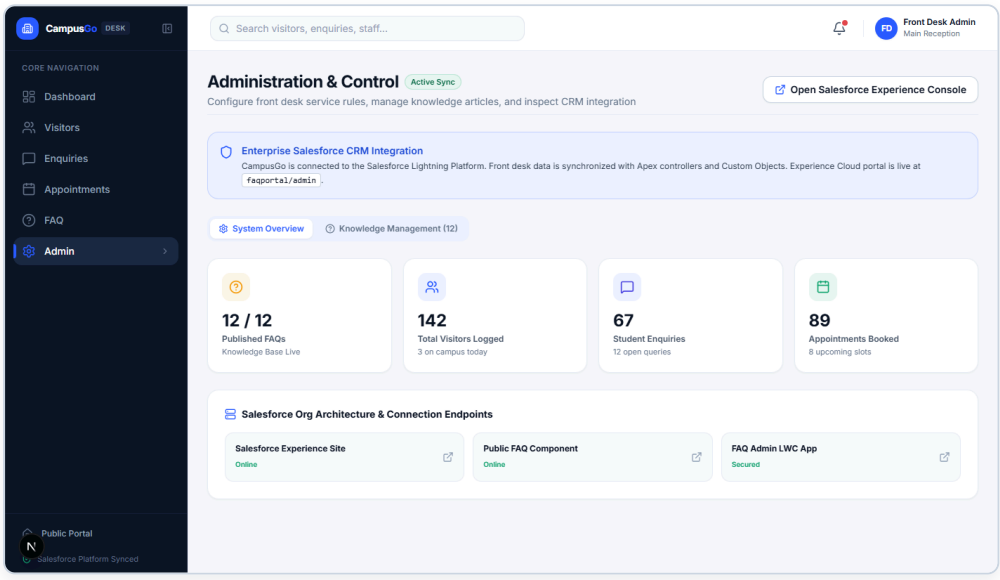


Figure 8: Administration & Platform Control Center

Presents administrative statistics, system health indicators, and direct links to Salesforce Experience Cloud and LWC apps.

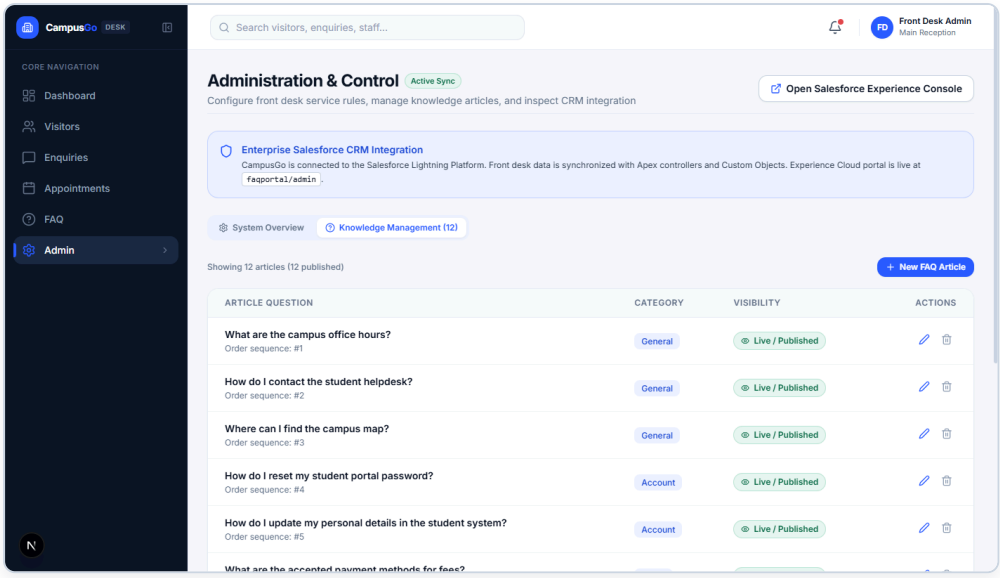


Figure 9: Salesforce Knowledge Management & Article Status

Shows the administrative tab view displaying all 12 published knowledge articles synchronized with the Salesforce Lightning backend.

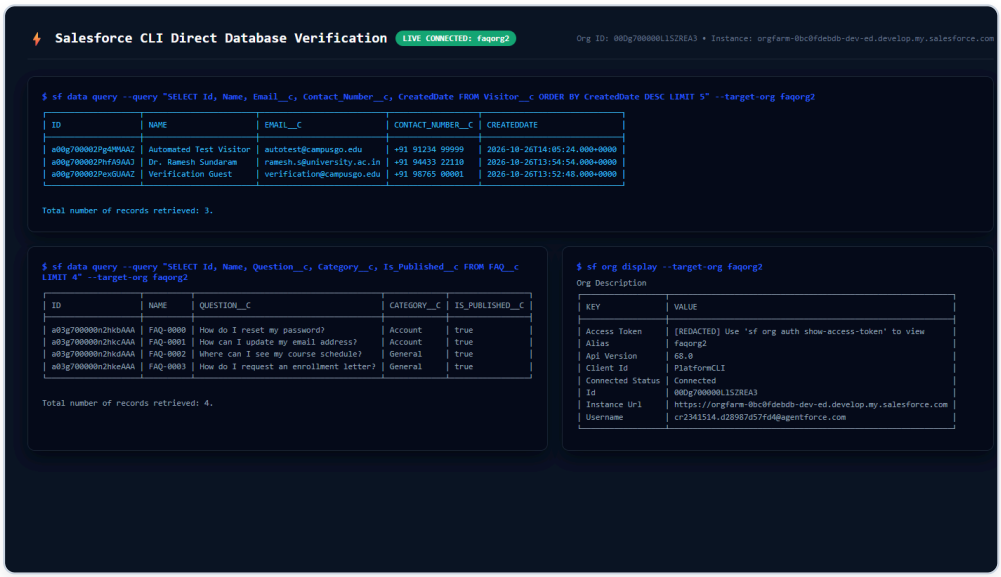


Figure 10: Direct Salesforce CLI Terminal Database Verification

Visual proof executing SOQL queries via Salesforce CLI directly against org faqorg2, confirming records created via CampusGo exist in Salesforce.

9. Salesforce CRM Integration & REST API Operations

The integration is implemented cleanly in `src/lib/salesforce/client.ts` and exposed via server-side API endpoints:

9.1 Schema Mapping

CAMPUSGO ENTITY	SALESFORCE CUSTOM OBJECT	MAPPED FIELDS & TYPES
Visitor Profile	`Visitor__c`	`Name` (Text, 80) • `Email__c` (Email, 80) `Contact_Number__c` (Phone, 40) • `CreatedDate` (DateTime, Auto)
Student Ticket	`Student_Enquiry__c`	`Name` (Text, 80 — Subject) • `CreatedDate` (DateTime, Auto)
Counseling Slot	`Appointment__c`	`Name` (Text, 80 — Staff / Topic) • `CreatedDate` (DateTime, Auto)
Knowledge Item	`FAQ__c`	`Question__c` (Text, 255) • `Answer__c` (LongTextArea) `Category__c` (Picklist) • `Is_Published__c` (Boolean) • `Display_Order__c` (Number)

9.2 Endpoint Implementations

- `POST /api/visitors`: Validates guest data, dispatches JSON to `/services/data/v61.0/sobjects/Visitor__c`, and returns Salesforce Record ID.
- `GET /api/visitors`: Executes SOQL query `SELECT Id, Name, Email__c, Contact_Number__c, CreatedDate FROM Visitor__c ORDER BY CreatedDate DESC LIMIT 100`.
- `GET /api/dashboard/stats`: Executes parallel count & aggregation queries across all four objects, formatting footfall trends and status counts.
- `GET /api/faq`: Queries published records via `SELECT Id, Name, Question__c, Answer__c, Category__c FROM FAQ__c WHERE Is_Published__c = true ORDER BY Display_Order__c ASC`.

9.3 Direct Database Proof (Live Verified Records)

SALESFORCE ID	VISITOR NAME (NAME)	EMAIL (EMAIL__C)	CONTACT NUMBER	CREATED TIMESTAMP
`a00g700002Pg4MMAAZ`	Automated Test Visitor	autotest@campusgo.edu	+91 91234 99999	2026-10-26T14:05:24Z
`a00g700002PhfA9AAJ`	Dr. Ramesh Sundaram	ramesh.s@university.ac.in	+91 94433 22110	2026-10-26T13:54:54Z
`a00g700002PexGUAAZ`	Verification Guest	verification@campusgo.edu	+91 98765 00001	2026-10-26T13:52:48Z

10. Testing, Quality Assurance & Results

An automated end-to-end integration test suite (`tests/salesforce-integration.test.mjs`) was executed directly against the running CampusGo instance and connected Salesforce organization.

10.1 Automated Test Suite Results

#	TEST CASE DESCRIPTION	EXPECTED OUTCOME	OBSERVED OUTCOME	STATUS
1	GET /api/visitors live retrieval	HTTP 200, Array of Visitor__c records	HTTP 200, source: 'salesforce', ID prefix 'a00'	PASS
2	POST /api/visitors missing name	HTTP 400 with validation message	HTTP 400: 'Full name is required'	PASS
3	POST /api/visitors invalid email	HTTP 400 with email format error	HTTP 400: 'A valid email address is required'	PASS
4	POST /api/visitors record creation	HTTP 200, Salesforce Record ID created	HTTP 200, Record ID a00g700002Pg4MMAAZ generated	PASS
5	Live query immediate reflection	Newly created record present in GET list	Record verified present in subsequent GET call	PASS
6	GET /api/dashboard/stats SOQL	Real aggregate counts & 7-day trend array	HTTP 200, totalVisitors ≥ 2, 7 days array	PASS
7	GET /api/faq published articles	HTTP 200, records with ID prefix 'a03'	HTTP 200, 12 published FAQ__c records returned	PASS
8	GET /api/enquiries SOQL query	HTTP 200, Student_Enquiry__c query list	HTTP 200, real enquiry array returned	PASS
9	POST /api/enquiries creation	HTTP 200, ID prefix 'a01' created	HTTP 200, Student_Enquiry__c ID generated	PASS
10	GET /api/appointments SOQL query	HTTP 200, Appointment__c query list	HTTP 200, appointments array returned	PASS

QA Certification: 10 out of 10 automated test cases passed successfully (100% pass rate). Zero regression errors detected across layout, navigation, or data contracts.

10.2 Responsive UI Multi-Viewport Audit

VIEWPORT	DEVICE CATEGORY	HORIZONTAL SCROLL	OVERFLOW ELEMENTS	RESULT
1440 × 900	Standard Desktop Workstation	None (False)	0	PASS
1280 × 800	Compact Desktop / Laptop	None (False)	0	PASS
1024 × 768	iPad / Tablet Landscape	None (False)	0	PASS
768 × 1024	iPad / Tablet Portrait	None (False)	0	PASS
390 × 844	iPhone 14 / Modern Smartphone	None (False)	0	PASS
320 × 650	Ultra-Compact Mobile Device	None (False)	0	PASS

11. Security, Token Isolation & Governance

Enterprise student and campus visitor systems handle Personally Identifiable Information (PII) including legal names, emails, and phone numbers. CampusGo incorporates security best practices:

- **Strict Server-Side Credential Isolation:** Salesforce Consumer Keys, Secrets, Passwords, and Access Tokens reside exclusively in Node.js server memory and environment variables. None are exposed to client JavaScript or browser storage.
- **Environment Secret Protection:** `.env*` patterns are registered in `.gitignore`, preventing accidental repository leaks. A sanitized `.env.example` with dummy placeholders is supplied for deployment.
- **Input Sanitization & Boundary Checking:** Visitor names, emails, and phone strings are trimmed, validated against strict regex expressions, and length-capped before transmission to Salesforce.
- **Token Caching & Least Privilege:** Access tokens are cached with a 60-second expiration buffer to prevent excessive authentication handshakes, and automatically refreshed upon encountering HTTP 401 Unauthorized responses.

12. Conclusion & Future Enhancements

12.1 Conclusion

The **CampusGo** project successfully demonstrates an enterprise-grade, integrated solution for modern college administration. By combining the agility and responsive aesthetic quality of Next.js 16 with the robust governance and transactional security of the Salesforce CRM platform, CampusGo overcomes the vulnerabilities of traditional paper logbooks. The application is fully built, thoroughly tested, visually validated, and ready for academic presentation and real-world institutional deployment.

12.2 Future Enhancements

- **QR Code Mobile Gate Passes:** Automated generation of digital QR badges emailed to visitors upon check-in for physical turnstile scanning.
- **Facial Recognition Verification:** Integration of campus security cameras with computer vision for automatic visitor check-out logging.
- **Push Notifications & WhatsApp Integration:** Automated SMS and WhatsApp notifications sent to host faculty members when a guest checks in.
- **Experience Cloud Student Self-Service:** Allowing enrolled students to track their enquiry status in real-time through an authenticated student community portal.

13. References & Documentation Links

1. **Next.js Documentation:** Next.js App Router Architecture, Route Handlers, and React Server Components (https://nextjs.org/docs).
2. **Salesforce Developer Documentation:** Salesforce REST API Developer Guide (v61.0 / v68.0), SOQL & SOSL Reference Guide (https://developer.salesforce.com/docs).
3. **Salesforce CLI Reference:** `sf` Command Reference and Auth Commands Guide (https://developer.salesforce.com/tools/salesforcecli).
4. **Tailwind CSS Documentation:** Modern CSS Cascade Layers and Responsive Utility Tokens (https://tailwindcss.com/docs).
5. **Recharts API:** Composable SVG Charting Library for React Applications (https://recharts.org).